

Amygdala Cell Types, Molecular Identity, and Transcriptomic Characterization

Table of Contents

Introduction/Background

Basal amygdala fear neurons and extinction neurons

BLA interneuron populations: NPY neurogliaform and calbindin-positive dendrite-targeting interneurons

Central amygdala large aspiny and small aspiny neurons

Paralamina nucleus immature excitatory neurons

Introduction/Background

The amygdala is not one uniform structure: the basolateral complex is mostly cortex-like and glutamatergic, while the central amygdala is more striatum-like and mainly GABAergic. This matters for the cell types in the rest of the review, because the named populations fall into distinct anatomical and transmitter-defined compartments that are now being refined by single-cell transcriptomic atlases. (8 sources)

A useful starting frame is that the amygdala contains many anatomically and molecularly distinct subregions—more than 13 in current summaries—and these are commonly grouped into the basolateral amygdala (BLA), central amygdala (CeA), medial amygdala, and intercalated cell clusters. These divisions can be separated by morphology, connectivity, physiology, and marker expression. ([Hu et al., 2022](#)) The basolateral complex, including lateral amygdala (LA) and BLA, is generally described as the main input side of the amygdala and has cortex-like features, with most neurons being glutamatergic projection neurons. By contrast, the central amygdala is the main output side and is striatum-like, with predominantly GABAergic projection neurons. ([Waclaw et al., 2010](#)) ([Chung et al., 2016](#))

Across reviews and quantitative studies, the same transmitter-level organization is consistent. The BLA is enriched in glutamatergic principal neurons that support associative learning, while the CeA is composed mainly of inhibitory neurons involved in emotional processing and expression. ([Chung et al., 2016](#)) More specifically, modern summaries place the BLA at about 80% glutamatergic principal neurons and about 20% GABAergic inhibitory neurons, whereas the CeA primarily contains GABAergic neurons. ([Hu et al., 2022](#)) Quantitative stereology in mouse gives a similar range, with most LA/BLA neurons glutamatergic and a minority GABAergic, and with the BA containing about 22% GABAergic neurons and the LA about 16%. ([Paul et al., 2025](#)) ([Vereczki et al., 2021](#)) This is also reflected in nucleus-level descriptions that state the majority of BLA neurons are spiny glutamatergic neurons, while CEI and CEm mainly contain GABAergic neurons. ([Sarowar et al., 2020](#))

This broad organization gives the scaffold for the specific “classical” cell types in the user’s question. Basal amygdala fear and extinction neurons are subsets of the glutamatergic principal-cell system in the BLA/BA, whereas BLA NPY neurogliaform cells and calbindin-positive dendrite-targeting interneurons belong to the smaller inhibitory compartment of the basolateral complex. ([Chung et al., 2016](#)) ([Hu et al., 2022](#)) ([Vereczki et al., 2021](#)) In the CeA, the classical large aspiny and small aspiny neurons fall within a region that is overwhelmingly GABAergic. ([Waclaw et al., 2010](#)) ([Sarowar et al., 2020](#)) ([Paul et al., 2025](#))

Transcriptomics now adds a second layer beyond classical anatomy and neurotransmitter type. Single-cell

RNA-seq studies in adult mouse amygdala have resolved large numbers of neuronal cell types—130 in one recent atlas—and mapped their spatial distributions while also identifying subsets that change transcriptionally after fear learning and retrieval. (Paul et al., 2025) (Hochgerner et al., 2023) This is especially important for the present topic because it means that older functional labels such as “fear neuron,” “extinction neuron,” “large aspiny neuron,” or “neurogliaform interneuron” can increasingly be linked to molecularly defined transcriptomic classes rather than only morphology, location, or physiology. At the same time, comparative work in primates suggests that some amygdala cell types are conserved across mammals, while glutamatergic neuron classes in the basolateral complex may show species-specific specializations. (Totty et al., 2025)

Evidence

(Hu et al., 2022)

New Insights into the Pivotal Role of the Amygdala in Inflammation-Related Depression and Anxiety Disorder

"The amygdala complex contains more than 13 subregions, which can be distinguished according to their neuronal morphology, unique connectivity, physiology, and expression of specific molecular markers. It is usually divided into the basolateral amygdala (BLA), central amygdala (CeA), medial amygdala (MeA), and intercalated cell clusters [78]"

"Developmentally, the BLA is a cortical-like structure consisting of 80% glutamatergic principal neurons and approximately 20% GABAergic inhibitory neurons (Carlsen et al., 1988)(Keifer et al., 2015). In contrast, CeA is a striatum-like structure that primarily comprises GABAergic inhibitory neurons (McDonald, 1982)."

(Waclaw et al., 2010)

Developmental origin of the neuronal subtypes that comprise the amygdalar fear circuit in the mouse

"The LA and BLA represent the input to the amygdala and exhibit cortex-like characteristics with a majority of glutamatergic projection neurons (Davis et al., 2000). Conversely, the CA contains striatum-like GABAergic projection neurons and represents the major output of the amygdala (McDonald, 1982)(Davis et al., 2000)."

(Chung et al., 2016)

The role of inositol 1,4,5-trisphosphate 3-kinase A in regulating emotional behavior and amygdala function

"The basolateral nucleus of the amygdala (BLA) is highly enriched in glutamatergic principal neurons and is required for associative learning. The central nucleus of the amygdala (CeA) primarily consists of GABAergic medium spiny neurons and controls the processing and expression of emotion."

(Paul et al., 2025)

Neuronal physiology of amygdala neurons in the context of injury and pain

"Most of the LA and BLA consists of glutamatergic excitatory neurons (80-85 %) (Vereczki et al., 2021). While the CeA consists of mostly inhibitory neurons (McDonald, 2003). A recent study reported that all the analyzed CeA neurons robustly clustered to GABA markers. This study incorporated single-cell transcriptomics to map the spatial distribution of amygdala cell types based on their neurotransmitter identity (Hochgerner et al., 2023)."

(Vereczki et al., 2021)

Total Number and Ratio of GABAergic Neuron Types in the Mouse Lateral and Basal Amygdala

"GABAergic neurons are key circuit elements in cortical networks. Despite growing evidence showing that inhibitory cells play a critical role in the lateral (LA) and basal (BA) amygdala functions, neither the number of GABAergic neurons nor the ratio of their distinct types has been determined in these amygdalar nuclei. Using unbiased stereology, we found that the ratio of GABAergic neurons in the BA (22%) is significantly higher than in the LA (16%) in both male and female mice. No difference was observed between the right and left hemispheres in either sex. In addition, we assessed the ratio of the major inhibitory cell types in both amygdalar nuclei. Using transgenic mice and a viral strategy for visualizing inhibitory cells combined with immunocytochemistry, we estimated that the following cell types together compose the vast majority of GABAergic cells in the LA and BA: axo-axonic cells (5.5%-6%), basket cells expressing parvalbumin (17%-20%) or cholecystokinin (7%-9%), dendrite-targeting inhibitory cells expressing somatostatin (10%-16%), NPY-containing neurogliaform cells (14%-15%), VIP and/or calretinin-expressing interneuron-selective interneurons (29%-38%), and GABAergic projection neurons expressing somatostatin and neuronal nitric oxide synthase (5.5%-8%). Our results show that these amygdalar nuclei contain all major GABAergic neuron types as found in other cortical regions. Furthermore, our data offer an essential reference for future studies aiming to reveal changes in GABAergic cell number and in inhibitory cell types typically observed under different pathologic conditions, and to model functioning amygdalar networks in health

and disease. **SIGNIFICANCE STATEMENT** GABAergic cells in cortical structures, as in the lateral and basal nucleus of the amygdala, have a determinant role in controlling circuit operation. In this study, we provide the first estimate for the total number of inhibitory cells in these two amygdalar nuclei. In addition, our study is the first to define the ratio of the major GABAergic cell types present in these cortical networks. Taking into account that hyperexcitability in the amygdala, arising from the imbalance between excitation and inhibition typifies many altered brain functions, including anxiety, post-traumatic stress disorder, schizophrenia, and autism, uncovering the number and ratio of distinct amygdalar inhibitory cell types offers a solid base for comparing the changes in inhibition in pathologic brain states."

(Sarowar et al., 2020)

[Rho GTPases in the Amygdala—A Switch for Fears?](#)

"The majority of BLA neurons are spiny glutamatergic neurons (with a minority of GABAergic interneurons) (Spampanato et al., 2011). CEI and CEm mainly contain GABAergic neurons."

(Hochgerner et al., 2023)

[Neuronal types in the mouse amygdala and their transcriptional response to fear conditioning](#)

"The neuronal subtypes regulating the emotional component of learning and memory are not fully described. Here the authors provide a molecular atlas of the adult mouse amygdala and show the subsets of cells transcriptionally responsive to fear. The amygdala is a brain region primarily associated with emotional response. The use of genetic markers and single-cell transcriptomics can provide insights into behavior-associated cell state changes. Here we present a detailed cell-type taxonomy of the adult mouse amygdala during fear learning and memory consolidation. We perform single-cell RNA sequencing on naïve and fear-conditioned mice, identify 130 neuronal cell types and validate their spatial distributions. A subset of all neuronal types is transcriptionally responsive to fear learning and memory retrieval. The activated engram cells upregulate activity-response genes and coordinate the expression of genes associated with neurite outgrowth, synaptic signaling, plasticity and development. We identify known and previously undescribed candidate genes responsive to fear learning. Our molecular atlas may be used to generate hypotheses to unveil the neuron types and neural circuits regulating the emotional component of learning and memory."

(Totty et al., 2025)

[Transcriptomic diversity of amygdalar subdivisions across humans and nonhuman primates](#)

"Compositional analyses revealed that subdivisions of the primate basolateral complex contain distinct classes of glutamatergic neurons and divergent gene expression profiles for parvalbumin and somatostatin GABAergic neurons. Referencing primate neuron types to transcriptomic atlases of the murine amygdala highlighted primate-specific specializations for glutamatergic neurons and identified which neuron types are conserved across mammals and amygdalar subdivisions."

Basal amygdala fear neurons and extinction neurons

In the basal/basolateral amygdala, "fear" and "extinction" neurons are usually treated as two distinct functional groups of glutamatergic principal neurons. Work combining activity-based labeling with RNA-seq then linked at least one fear-suppressing or "Fear-Off" population to markers such as **Ntsr2**, **Dkk3**, **Rspo2**, and **Wnt7a**. (2 sources)

Within BA/BLA principal cells, classical fear and extinction neurons are defined first by what they do during conditioning and extinction, rather than by one single canonical molecular marker. Fear neurons are activated by the conditioned stimulus during fear expression, whereas extinction neurons become active after repeated conditioned-stimulus presentations without the unconditioned stimulus, making them a functionally distinct and non-overlapping population in this framework. Both populations project to medial prefrontal cortex, but only extinction neurons receive reciprocal mPFC input, which gives a clear circuit basis for top-down control of extinction-related activity. ([Cardenas et al., 2019](#))

In neurotransmitter terms, these fear and extinction populations sit within the BLA principal-cell compartment and are therefore best understood as excitatory, glutamatergic projection neurons rather than local inhibitory interneurons. Their "molecular identity" has historically been weaker than their circuit and behavioral identity, because the original definitions came from activity during fear learning and extinction rather than from transcriptomics alone. ([Cardenas et al., 2019](#)) (Model-Generated)

A key step toward molecularly anchoring these classical populations came from activity-dependent and genetically targeted labeling studies of behaviorally defined BLA neurons. McCullough and colleagues used Thy1-based labeling to isolate a BLA population that consistently behaved as fear-inhibiting or "Fear-Off" neurons, then applied RNA sequencing to identify enriched transcripts. In that population, **Ntsr2**, **Dkk3**,

Rspo2, and **Wnt7a** were strongly upregulated, and follow-up experiments highlighted **Ntsr2**-expressing BLA neurons as a putative Fear-Off cell class. This is one of the clearest examples of linking a classical extinction- or fear-suppressing phenotype to candidate transcriptomic markers. ([McCullough et al., 2016](#))

So, for the user's query about labeling and transcriptomic studies: the older "fear neuron" versus "extinction neuron" split is mainly a functional and projection-defined scheme, while later activity-guided RNA profiling began to assign marker genes to fear-suppressing BA/BLA cells. In the references provided here, the strongest transcriptomic handle is on the Fear-Off/extinction-like side, especially **Ntsr2** and the associated enriched genes **Dkk3**, **Rspo2**, and **Wnt7a**. ([Cardenas et al., 2019](#)) ([McCullough et al., 2016](#))

Evidence

(Cardenas et al., 2019)

[Persistent pain intensifies recall of consolidated fear memories](#)

"With respect to fear expression, BLA principal neurons can be divided into two functionally distinct, non-overlapping populations. Activation of "fear" neurons is triggered by the conditioned stimulus, while "extinction" neurons become active only after repetitive presentations of the conditioned stimulus that are not followed by the unconditioned stimulus (Herry et al., 2008). Both types of neurons project to the mPFC but only extinction neurons receive reciprocal input from the mPFC, which makes their activity susceptible to mPFC modulation (Herry et al., 2008)."

(McCullough et al., 2016)

[Molecular characterization of Thy1 expressing fear-inhibiting neurons within the basolateral amygdala](#)

"Here we demonstrate a comprehensive workflow for identification of pharmacologically tractable markers of behaviourally characterized cell populations. Thy1-eNpHR-, Thy1-Cre- and Thy1-eYFP-labelled neurons of the BLA consistently act as fear inhibiting or 'Fear-Off' neurons during behaviour"

".RNA sequencing identifies genes strongly upregulated in RNA of this population, including Ntsr2, Dkk3, Rspo2 and Wnt7a"

".These experiments identify and validate Ntsr2-expressing neurons within the BLA, as a putative 'Fear-Off' population."

BLA interneuron populations: NPY neurogliaform and calbindin-positive dendrite-targeting interneurons

*In the BLA, both NPY neurogliaform cells and calbindin-positive dendrite-targeting cells belong to the local **GABAergic interneuron** pool, not the glutamatergic principal-cell pool. In the material provided here, only the **calbindin-positive** class is directly tied to an activity-dependent study, while no patch-seq or single-cell RNA-seq marker set for either class is supplied in the quoted references. (1 source)*

The two classical BLA interneuron types in the query are best placed within the inhibitory minority of the basolateral complex. **NPY neurogliaform interneurons** are classically understood as local **GABAergic** cells with neurogliaform-like morphology and **NPY** expression, while **calbindin-positive dendrite-targeting interneurons** are likewise inhibitory **GABAergic** interneurons defined by **calbindin (CALB1/CB)** immunoreactivity and dendrite-focused innervation patterns rather than by long-range projection identity. Their "molecular identity" in the older literature is therefore mostly marker-based and morpho-physiological—**NPY** for the neurogliaform class, and **calbindin** for the dendrite-targeting class—rather than grounded in a single modern transcriptomic cluster name. (Model-Generated)

Among the studies provided for this section, the clearest direct characterization concerns the **calbindin-positive BLA interneuron** population in an activity-dependent setting. During fear extinction, pharmacological blockade of orexin-1 receptors increased activity in infralimbic-projecting BLA neurons and also activated **GABAergic interneurons expressing calbindin, but not parvalbumin**, in the BLA. This supports the idea that calbindin-positive interneurons are a distinct inhibitory BLA cell class recruited during extinction-related network states. ([Flores et al., 2017](#))

For the user's specific question about activity-dependent labeling, patch-seq, and single-cell RNA-seq, the

evidence in the supplied quotes is uneven. The provided reference supports **activity-dependent recruitment of calbindin-positive GABAergic interneurons** during extinction-related manipulations, but it does not give a patch-seq profile or a full scRNA-seq marker panel for this class, and it provides **no direct transcriptomic dataset** for **BLA NPY neurogliaform interneurons**. So, based on the references given here, these two BLA interneuron populations remain best described at the classical level as **GABAergic inhibitory interneurons** marked by **NPY** or **calbindin**, with direct activity-linked evidence presently stronger for the **calbindin-positive** population than for the **NPY neurogliaform** population. ([Flores et al., 2017](#)) (Model-Generated)

Evidence

(Flores et al., 2017)

[Facilitation of Contextual Fear Extinction by Orexin-1 Receptor Antagonism Is Associated with the Activation of Specific Amygdala Cell Subpopulations](#)

"Treatment with the orexin-1 receptor antagonist SB334867 increased the activity of basolateral amygdala neurons projecting to infralimbic medial prefrontal cortex during fear extinction. GABAergic interneurons expressing calbindin, but not parvalbumin, were also activated by orexin-1 receptor antagonism in the basolateral amygdala."

Central amygdala large aspiny and small aspiny neurons

*The classical central amygdala "large aspiny" and "small aspiny" neurons should be placed first in a ****CeA-wide GABAergic framework****. In the references provided here, their old morphology-based names are supported much better than any one modern transcriptomic marker set, and newer single-cell work mainly shows that CeA neurons cluster strongly with ****GABAergic identity****. (6 sources)*

For the two classical CeA cell types in the query, the most secure conclusion from the supplied papers is about **region and transmitter identity** rather than a precise modern marker panel. The central amygdala is consistently described as a **striatum-like amygdala subdivision** and the main output side of the complex, in contrast to the cortex-like BLA. Its neurons are therefore understood primarily as **GABAergic projection neurons** rather than glutamatergic principal cells. This basic point is repeated across developmental, systems, and review papers. ([Waclaw et al., 2010](#)) ([Hu et al., 2022](#)) ([Chung et al., 2016](#)) ([Sarowar et al., 2020](#))

Within that framework, the older terms "**large aspiny neuron**" and "**small aspiny neuron**" are best read as **classical morphological classes of CeA inhibitory neurons**. The provided references do not assign either class to a unique gene-defined transcriptomic type, nor do they give a direct patch-seq dataset for them. So their molecular identity remains limited here: both belong to a CeA compartment that is mainly **GABAergic**, and their distinction is still primarily **size- and morphology-based** in the context of these sources. ([Waclaw et al., 2010](#)) ([Hu et al., 2022](#))

The newer transcriptomic evidence in the supplied set strengthens this transmitter-level interpretation but does not cleanly recover the old "**large aspiny**" versus "**small aspiny**" labels. A recent summary notes that single-cell transcriptomic mapping found analyzed CeA neurons to cluster robustly with **GABA markers**, and points to the adult mouse amygdala atlas of Hochgerner and colleagues. That atlas identified **130 neuronal cell types** across the amygdala and mapped their spatial distributions, providing a modern framework in which classical CeA morphologies can eventually be connected to molecular classes. But from the quotes given here, the key takeaway is still that **CeA cell types, including the classical aspiny neurons, are overwhelmingly inhibitory/GABAergic**. ([Paul et al., 2025](#)) ([Hochgerner et al., 2023](#))

For the user's question about **activity-dependent labeling, patch-seq, and single-cell RNA-seq studies**, the evidence supplied for these CeA populations is narrow. In this reference set, there is **no direct activity-dependent labeling study** tied specifically to classical **large aspiny** or **small aspiny** neurons, and **no patch-seq study** is provided for them. The relevant modern study type here is **single-cell RNA-seq**, which supports the broader conclusion that CeA neurons are **GABAergic** and supplies a transcriptomic atlas of amygdala cell types, but it does not, in the quoted material, name a one-to-one transcriptomic marker set for the classical large and small aspiny categories. ([Paul et al., 2025](#)) ([Hochgerner et al., 2023](#))

Evidence

(Waclaw et al., 2010)

Developmental origin of the neuronal subtypes that comprise the amygdalar fear circuit in the mouse

"The LA and BLA represent the input to the amygdala and exhibit cortex-like characteristics with a majority of glutamatergic projection neurons (Davis et al., 2000). Conversely, the CA contains striatum-like GABAergic projection neurons and represents the major output of the amygdala (McDonald, 1982)(Davis et al., 2000)."

(Hu et al., 2022)

New Insights into the Pivotal Role of the Amygdala in Inflammation-Related Depression and Anxiety Disorder

"The amygdala complex contains more than 13 subregions, which can be distinguished according to their neuronal morphology, unique connectivity, physiology, and expression of specific molecular markers. It is usually divided into the basolateral amygdala (BLA), central amygdala (CeA), medial amygdala (MeA), and intercalated cell clusters [78]"

".Developmentally, the BLA is a cortical-like structure consisting of 80% glutamatergic principal neurons and approximately 20% GABAergic inhibitory neurons (Carlsen et al., 1988)(Keifer et al., 2015). In contrast, CeA is a striatum-like structure that primarily comprises GABAergic inhibitory neurons (McDonald, 1982)."

(Chung et al., 2016)

The role of inositol 1,4,5-trisphosphate 3-kinase A in regulating emotional behavior and amygdala function

"The basolateral nucleus of the amygdala (BLA) is highly enriched in glutamatergic principal neurons and is required for associative learning. The central nucleus of the amygdala (CeA) primarily consists of GABAergic medium spiny neurons and controls the processing and expression of emotion."

(Sarowar et al., 2020)

Rho GTPases in the Amygdala—A Switch for Fears?

"The majority of BLA neurons are spiny glutamatergic neurons (with a minority of GABAergic interneurons) (Spampanato et al., 2011). CEI and CEm mainly contain GABAergic neurons."

(Paul et al., 2025)

Neuronal physiology of amygdala neurons in the context of injury and pain

"Most of the LA and BLA consists of glutamatergic excitatory neurons (80-85 %) (Vereczki et al., 2021). While the CeA consists of mostly inhibitory neurons (McDonald, 2003). A recent study reported that all the analyzed CeA neurons robustly clustered to GABA markers. This study incorporated single-cell transcriptomics to map the spatial distribution of amygdala cell types based on their neurotransmitter identity (Hochgerner et al., 2023)."

(Hochgerner et al., 2023)

Neuronal types in the mouse amygdala and their transcriptional response to fear conditioning

"The neuronal subtypes regulating the emotional component of learning and memory are not fully described. Here the authors provide a molecular atlas of the adult mouse amygdala and show the subsets of cells transcriptionally responsive to fear. The amygdala is a brain region primarily associated with emotional response. The use of genetic markers and single-cell transcriptomics can provide insights into behavior-associated cell state changes. Here we present a detailed cell-type taxonomy of the adult mouse amygdala during fear learning and memory consolidation. We perform single-cell RNA sequencing on naïve and fear-conditioned mice, identify 130 neuronal cell types and validate their spatial distributions. A subset of all neuronal types is transcriptionally responsive to fear learning and memory retrieval. The activated engram cells upregulate activity-response genes and coordinate the expression of genes associated with neurite outgrowth, synaptic signaling, plasticity and development. We identify known and previously undescribed candidate genes responsive to fear learning. Our molecular atlas may be used to generate hypotheses to unveil the neuron types and neural circuits regulating the emotional component of learning and memory."

Paralaminar nucleus immature excitatory neurons

*The paralaminar nucleus (PL) is notable for a long-lasting pool of ****immature excitatory neurons**** rather than a mature inhibitory cell class. In the reference provided here, their key identity is developmental: they are excitatory neurons that stay immature across childhood and adolescence, with some evidence that local progenitors may add to this pool early in life. (1 source)*

In the context of the user's query, the **molecular identity** of paralaminar nucleus cells is best stated cautiously from the supplied evidence: they are a **population of immature excitatory neurons**, not the mainly GABAergic classes seen in the central amygdala and not the mature glutamatergic principal neurons usually emphasized in the BLA. Page and colleagues describe the PL as containing "a large population of immature excitatory neurons at birth," with the added possibility that "some of which may continue to be born from local progenitors." Those progenitors then "disappear rapidly in infancy," while the immature neurons themselves persist through childhood and adolescence, implying a **protracted developmental program** rather than a fast transition into a fully mature adult-like state. ([Page et al., 2022](#))

So, for **neurotransmitter type**, the safest answer from these quotes is **excitatory**, and for **transcriptomic or marker identity**, the present reference supports an **immature-neuron developmental identity** more than a specific gene panel. In other words, this is one of the queried amygdala populations whose defining feature in the supplied literature is **developmental state** as much as region or morphology. ([Page et al., 2022](#))

For the user's last question about **activity-dependent labeling, patch-seq, and single-cell RNA-seq studies**, the material provided for this section is limited. The cited study is best used here as a **developmental and molecular-state characterization** of PL neurons, not as an activity-labeling or patch-seq paper, and the quote supplied does **not** provide a specific single-cell RNA-seq marker set for this population. Based on the evidence given here, the clearest conclusion is that the paralaminar nucleus contains a **persistent pool of immature excitatory neurons** with a prolonged maturation timeline. ([Page et al., 2022](#))

Evidence

(Page et al., 2022)

[Immature excitatory neurons in the amygdala come of age during puberty](#)

"The PL contains a large population of immature excitatory neurons at birth, some of which may continue to be born from local progenitors. These progenitors disappear rapidly in infancy, but the immature neurons persist throughout childhood and adolescent ages, indicating that they develop on a protracted timeline."